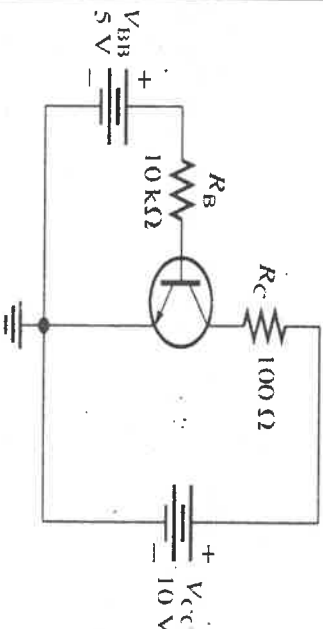
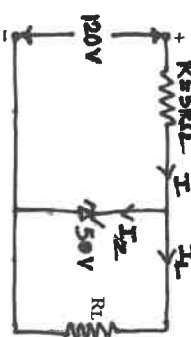


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II SEMESTER B.Tech. MID TERM EXAMINATION, FEB.-2026 Paper Code – EC/EE/IC ECC02, EE ECC02 Subject: Electronic Devices & Circuits Time: 1.5 Hours Max. Marks – 15 Note: - Missing data/ information if any, may be suitably assumed & mentioned in the answer. All questions carry equal marks [3=1, 5+1, 5]			
Question	CO		
1a Determine I_B, I_C, I_E, V_{CB}, and V_{CE} in the circuit and region of operation. Assume that: $\beta = 150$ and $V_{BE} = 0.7V$. 	CO ₁		
1b Draw and explain input and output V-I characteristics of common emitter (CE) configuration of BJT. Also derive the relationship between β and α .	CO ₁		
2a Derive the expression for the contact barrier potential (built-in potential) of a step-graded PN junction. State all assumptions clearly.	CO ₁		
2b Draw and explain Ebers-Moll model of a BJT.	CO ₂		
3.a In electronics circuit a germanium PN junction diode is used which carries a current of 10.25 mA when it is forward biased with 0.25 V at $T = 300K$. Find – I. Reverse Saturation Current. II. Bias Voltage required for getting a current of 120 mA.	CO ₂		

3.b For the given Zener diode regulator circuit, determine the load voltage V_L , load current I_L , Zener current I_Z , and series resistor current I_R when the load resistance $R_L = 1 k\Omega$. Also, determine the minimum value of R_L required to ensure that the Zener diode operates in the breakdown (ON) region.	CO ₄
	
4a Compare CB, CC and CE configuration of a BJT.	CO ₁
4b What is operating point? Explain any two biasing schemes with help of suitable circuit diagram.	CO ₃
5a Show Fermi levels in extrinsic and intrinsic semiconductor. Also, calculate the hole concentration in a silicon crystal with a donor concentration of $1.4 \times 10^{24} m^{-3}$, given that the intrinsic carrier concentration is $1.4 \times 10^{18} m^{-3}$. Further, determine the ratio of electron concentration to hole concentration.	CO ₃
5b The mobilities of free electrons and holes in pure germanium are 3800 $cm^2/V \cdot s$ and 1800 $cm^2/V \cdot s$, respectively. The corresponding mobilities in pure silicon are 1300 $cm^2/V \cdot s$ and 500 $cm^2/V \cdot s$, respectively. Assuming the intrinsic carrier concentration at room temperature to be $2.5 \times 10^{13} cm^{-3}$ for germanium and $1.5 \times 10^{10} cm^{-3}$ for silicon, calculate the intrinsic electrical conductivity of both materials.	CO ₄